

Geospatial Analysis of Electoral Outcomes in Trento: The 2025 Ianeselli Case

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Abstract

This paper investigates the socio-demographic and spatial drivers of voting behavior during the May 2025 municipal elections in Trento, Italy, which saw the confirmation of the center-left incumbent mayor, Franco Ianeselli. While traditional ecological inference models often assume spatial independence, this study applies geospatial analysis techniques to precinct-level data to account for geographical proximity and social contagion. By integrating electoral results with ISTAT census data and open transport data, we estimate a Spatial Autoregressive model (SAR). The results reveal a strong, positive spatial autocorrelation in progressive voting. Furthermore, we demonstrate that while factors like population density and non-EU resident concentration appear significant in standard non-spatial OLS regressions, their effects disappear when controlling for spatial spillovers. Ultimately, the educational level of the electorate emerges as the only robust direct driver of the progressive vote, generating significant positive spatial spillovers on neighboring precincts. These findings highlight the necessity of spatial modeling in political geography to avoid spurious correlations and better understand center-periphery electoral polarization.

Keywords: spatial autocorrelation, spatial autoregressive model (SAR), electoral data, electoral geography.

1. Research Question

The spatial dimension of electoral behavior is a crucial, yet often overlooked, component in understanding modern political dynamics. While traditional political science heavily relies on individual-level survey data or spatially-blind ecological regressions, voters are inherently embedded in geographic and social contexts. Following Tobler's First Law of Geography: "everything is related to everything else, but near things are more related than distant things" (Tobler (1970)), this paper aims to analyze the spatial patterns of the progressive vote in a mid-sized European urban context.

We focus on the municipal elections held in Trento (Italy) in May 2025. The election resulted in the first-round confirmation of the incumbent center-left mayor, Franco Ianeselli, who secured over 54.65% of the votes, comfortably defeating the center-right candidate, Ilaria Goio (26,60%). However, this decisive victory was accompanied by a historically low voter turnout (49.93%), signaling underlying political disengagement and localized discontent.

A purely aggregate analysis masks profound intra-urban polarization. The 2025 electoral campaign was heavily characterized by debates over major infrastructural projects, most

notably the construction of the *Bypass Ferroviario* (railway bypass). The localized impact of these construction sites provides a unique lens through which to observe center-periphery dynamics and NIMBY (Not In My Back Yard) electoral penalties. Interestingly, preliminary observations at the precinct level reveal a highly nuanced spatial response. While precincts immediately adjacent to the northern construction sites (specifically sections 11, 12, and 14) visibly suffered from localized discontent, dropping below the 50% absolute majority threshold, the broader northern district absorbed this shock without a generalized collapse. Conversely, in the southern suburb of Mattarello, the infrastructural disruption triggered a severe, district-wide electoral penalty. This geographic variation suggests that localized infrastructural grievances do not operate in a vacuum, but rather, they interact directly with the underlying socio-demographic fabric and spatial isolation of the neighborhoods, requiring a multivariate spatial approach to be fully disentangled.

To systematically investigate these dynamics, this study addresses the following Research Questions (RQs):

- **RQ1 (Socio-demographic drivers):** To what extent do socio-demographic factors, particularly higher education and urbanization metrics, drive support for the progressive incumbent at the precinct level?
- **RQ2 (Spatial Contagion):** Is electoral behavior purely compositional, or do spatial spillover effects (social contagion and neighborhood interactions) significantly influence voting patterns across neighboring precincts?
- **RQ3 (Methodological Bias):** How does the inclusion of spatial dependence alter the perceived significance of urban density and immigration in explaining voting behavior compared to traditional non-spatial models?

To answer these questions, we construct a novel dataset combining precinct-level electoral results ([Napolitano \(2025\)](#)), ISTAT census data ([Istituto Nazionale di Statistica \(ISTAT\) \(2021\)](#)), and GTFS public transport data ([Open Data Trentino \(2025\)](#)). Overcoming the Modifiable Areal Unit Problem (MAUP) and boundary mismatch issues typical of Italian electoral geography ([Pinto \(2023\)](#)), we employ a spatial econometric framework, ultimately selecting a Spatial Autoregressive (SAR) model.

The remainder of this paper is structured as follows. Section 2 reviews the related work on spatial electoral geography. Section 3 describes the dataset construction and the spatial methodology. Section 4 presents the exploratory data analysis and tests for spatial autocorrelation. Section 5 details the spatial econometric models and discusses the impacts. Finally, Section 6 provides the code and concluding remarks.

2. Related Work

Building upon the theoretical premise that spatial dependence fundamentally shapes electoral outcomes, contemporary political geography has increasingly adopted spatial econometric techniques to untangle the complex drivers of urban voting behavior. The transition from spatially-blind methodologies to models that explicitly account for neighborhood interactions is central to recent literature, allowing researchers to isolate the true impact of local socio-demographic contexts.

2.1. Socio-Demographic Drivers and Urban Centrality

Recent geospatial analyses have extensively investigated the socio-demographic drivers of voting behavior. While a significant portion of this literature has focused on identifying the spatial determinants of populist or radical right support (Sümeghy (2024); Štefan Rehák and Kuběnková (2026)), the underlying contextual variables validated by these studies are equally fundamental for analyzing the progressive vote. Structural variables such as the concentration of immigrants, education levels, and urban density are consistently recognized as primary predictors of spatial political polarization.

In our analysis, we utilize the concentration of non-EU residents, population density, the share of highly educated individuals, and the proximity to public transit to capture the socio-economic fabric of Trento. High education levels and urban centrality, are traditionally associated with progressive voting patterns. Conversely, peripheral areas tend to exhibit different electoral behaviors due to relative marginalization. By applying these established variables to the incumbent center-left candidate, we aim to validate their role as drivers for the progressive vote. Additionally, while structural variables explain general center-periphery divides, localized events such as the construction of major infrastructural projects can generate localized discontent or NIMBY (Not In My Back Yard) effects, slightly altering the expected spatial voting patterns in specific neighborhoods.

2.2. Ecological Fallacy and the Need for Granular Data

A purely aggregate analysis at the municipal level masks profound intra-urban polarization and risks incurring the ecological fallacy, the logical error of deducing individual or neighborhood-level behavior solely from aggregate statistics. Analyzing electoral results at the most granular level available, the precinct (*sezione elettorale*), is therefore essential to capture true spatial dynamics. However, obtaining geocoded electoral results at the precinct level has historically been a major challenge in Italy due to the lack of standardized spatial boundaries for electoral sections. Pinto (2023) highlights the absolute necessity of overcoming these data limitations to accurately observe spatial inequalities within cities. In line with this methodological imperative, our study performs a complex spatial aggregation of census blocks and electoral precincts to properly map the voting behavior.

2.3. The Methodological Shift to Spatial Regression Models

From a methodological standpoint, the literature has increasingly recognized the limitations of traditional non-spatial models. Ordinary Least Squares (OLS) regressions assume spatial independence among observations. However, geographic proximity and social contagion inevitably lead to spatial autocorrelation, violating the fundamental assumption of independence (Anselin (1995)). Ignoring this spatial dependence produces biased and inconsistent estimates.

To properly address spatial autocorrelation and quantify how voting behaviors spill over across neighboring precincts, the literature advocates for the transition from OLS to spatial regression frameworks (LeSage and Pace (2009)). Relying on the model selection strategies formalized by Elhorst (2010), this study shifts the focus towards Spatial Autoregressive (SAR) and Spatial Durbin Models (SDM). These models allow us not only to evaluate the direct impact of a neighborhood's characteristics on its own voting outcomes, but also to measure

the indirect impacts (global or local spatial spillovers), revealing how the socio-demographic composition of neighboring areas influences local progressive support.

3. Description of the Data

To investigate the spatial determinants of electoral behavior, this study integrates three distinct datasets: official electoral results from the Municipality of Trento, socio-demographic data from the 2021 ISTAT Census, and Trento public transport open data (GTFS).

3.1. Study Area and Spatial Aggregation (The MAUP Challenge)

The empirical analysis covers the urban territory of Trento, Italy. The chosen unit of analysis is the standard electoral precinct (*sezione elettorale*). After systematically excluding special, military, and flying precincts (e.g., precincts located in hospitals or specific institutions), the final dataset yields $N = 80$ spatial units. Integrating census data with electoral boundaries presents a methodological challenge known as the Modifiable Areal Unit Problem (MAUP), due to the spatial mismatch between ISTAT census blocks and electoral precincts. To resolve this, a spatial join process was implemented using areal interpolation by majority rule. Census variables were assigned to the electoral precinct that physically contained the majority of the census block's residential buildings. While acknowledging the inherent limitations of this approximation, this technique ensures statistical robustness while preserving micro-level granularity, an approach extensively validated in recent literature concerning Italian electoral data (Pinto (2023)).

3.2. Dependent Variable: Electoral Vote Share

The dependent variable captures the localized electoral preferences during the most recent municipal elections.

- **Vote Share (*ianeselli_vote_pct*):** Defined as the percentage of valid votes cast for the mayoral candidate Ianeselli over the total valid votes recorded within each electoral precinct. This continuous variable serves to identify spatial clusters of political support across different neighborhoods.

3.3. Independent Variables: Socio-Demographic Context and Transport

To isolate the structural determinants of voting behavior, we include a set of socio-economic controls and a transport accessibility metric.

- **Education (*highly_educated_pct*):** The proportion of residents holding a university degree, acting as a proxy for the human capital and cultural orientation of the precinct.
- **Immigration (*extra_ue_pct*):** The percentage of non-EU residents.
- **Urban Density (*density*):** The population density per square kilometer, used to control for the urban-suburban divide.

- **Transit Accessibility (`nearest_transport_distance_m`):** An engineered spatial metric derived from GTFS data. It measures the average minimum walking distance (in meters) from residential buildings within the precinct to the nearest public transit stop, serving as a proxy for neighborhood connectivity and municipal service provision.

3.4. The Spatial Weights Matrix (W)

A critical component of spatial regression modeling is the definition of the spatial weights matrix (W), which formalizes the geographical connectivity between the spatial units. To ensure accurate distance-based measurements, spatial data were projected using the Universal Transverse Mercator (UTM) coordinate system, Zone 32N (EPSG:32632), based on the WGS84 datum.

Given the heterogeneous morphology of the municipality of Trento, defining neighborhood relationships requires careful consideration. A simple contiguity matrix (such as Queen or Rook) or a purely distance-based matrix would be inappropriate: peripheral and geographically isolated suburbs (such as Sardagna, Mattarello, Meano, and Povo) would exhibit very few connections, whereas the highly concentrated precincts in the city center would be disproportionately dense. Consequently, to ensure that each spatial unit has exactly the same number of neighbors regardless of its urban or suburban nature, we adopted a k -Nearest Neighbors (KNN) matrix.

- **Primary Matrix ($k = 4$):** The baseline model relies on $k = 4$ nearest neighbors to ensure baseline connectivity without oversmoothing local variations.
- **Row-Standardization:** The matrix was row-standardized. This ensures that the spatial lag operates as an unweighted average of the neighbors' values. In the context of electoral geography, this approach assumes that the neighborhood effect (spillover) is distributed equally among the k closest neighbors, effectively capturing the average surrounding socio-political environment rather than weighting strictly by metric proximity.
- **Robustness Checks:** To ensure that the empirical findings are not sensitive to the choice of the spatial topology, alternative weight matrices with $k = 5$ and $k = 6$ were tested. The results of these sensitivity analyses are reported in the Appendix.

4. Data Analysis

The empirical analysis is structured in progressive steps. First, an Exploratory Data Analysis (EDA) is conducted to assess the distribution of the dependent variable and its bivariate relationships with the socio-demographic covariates. Second, an Exploratory Spatial Data Analysis (ESDA) is performed to formally test and map global and local spatial autocorrelation. Finally, spatial regression models are estimated to quantify the structural determinants of the vote while controlling for spatial spillovers.

4.1. Exploratory Data Analysis (EDA)

Before introducing spatial weights, it is crucial to understand the baseline distribution of the mayoral vote share and its geographical polarization.

Univariate Analysis: Vote Distribution

Figure 1 illustrates the distribution of the vote share for the incumbent mayor, Ianeselli, across the $N = 80$ electoral precincts. The distribution is slightly right-skewed, with the mean and median values tightly aligned just above the absolute majority threshold (approximately 53-54%). While the overall city average confirms a solid victory, the variance across precincts is notable, with values ranging from a minimum of approximately 44% to peaks exceeding 70%. The concentration of sections in the 45-55% range highlights a high degree of competitiveness in a large portion of the city.

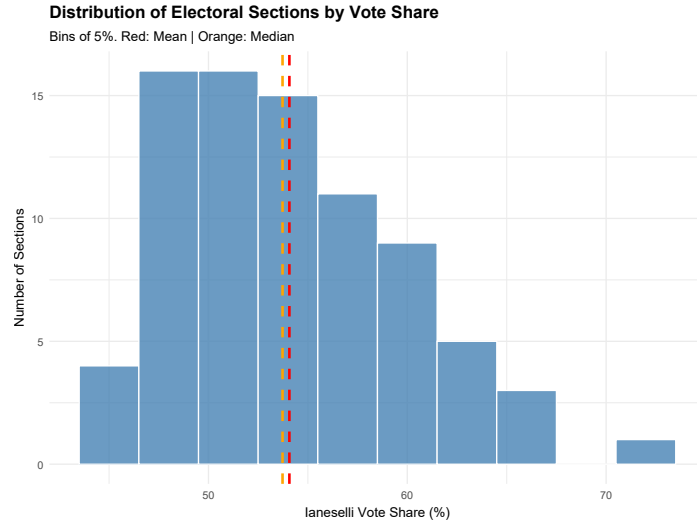


Figure 1: Distribution of Votes for Ianeselli across Trento's electoral precincts.

Spatial Polarization: The Urban-Suburban Divide

A deeper look at the precinct level reveals a distinct center-periphery divide. Table 1 highlights the extremes of this electoral distribution. The highest electoral support (Top 5 precincts) is entirely concentrated in the historical and central districts of the city, specifically *Oltrefersina* and *S. Giuseppe-S. Chiara*. Conversely, the lowest performance (Flop 5 precincts) is heavily localized in peripheral and suburban areas, almost exclusively in the *Mattarello* district.

However, it is important to contextualize this variance: although the center-periphery gap is significant, the incumbent mayor's performance remains remarkably strong across the board. As depicted in the boxplot (Figure 3), the median vote share for the vast majority of the municipal districts stays well above the 50% absolute majority threshold. This indicates that the spatial polarization represents a fluctuation within a widespread overall consensus, rather than a sharp rejection by the suburbs.

This spatial polarization is clearly depicted in the descriptive choropleth map (Figure 2) and further confirmed by the boxplot (Figure 3). The map employs a divergent color palette centered precisely at the 50% midpoint, intuitively separating the core urban precincts where the candidate secured an absolute majority from the surrounding suburban valleys where he struggled.

Bivariate Relationships

To preliminarily explore the structural determinants of the vote, a grid of bivariate scatterplots (Figure 4) was produced. The analysis reveals significant socio-demographic associations:

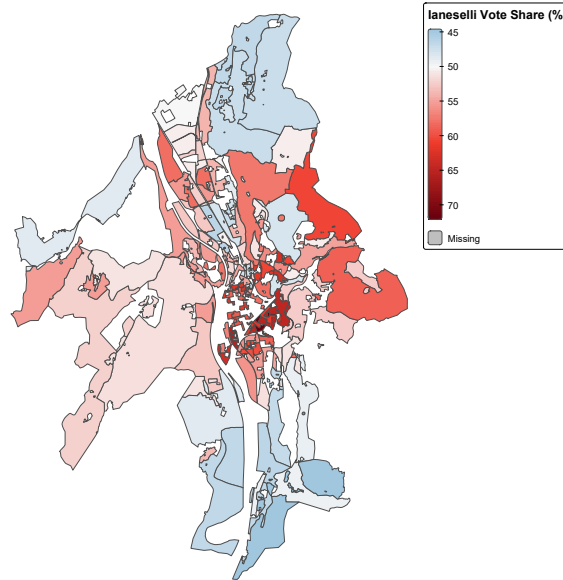
Electoral Geography: Vote Share by Section

Figure 2: Electoral Geography: Vote Share by Section.

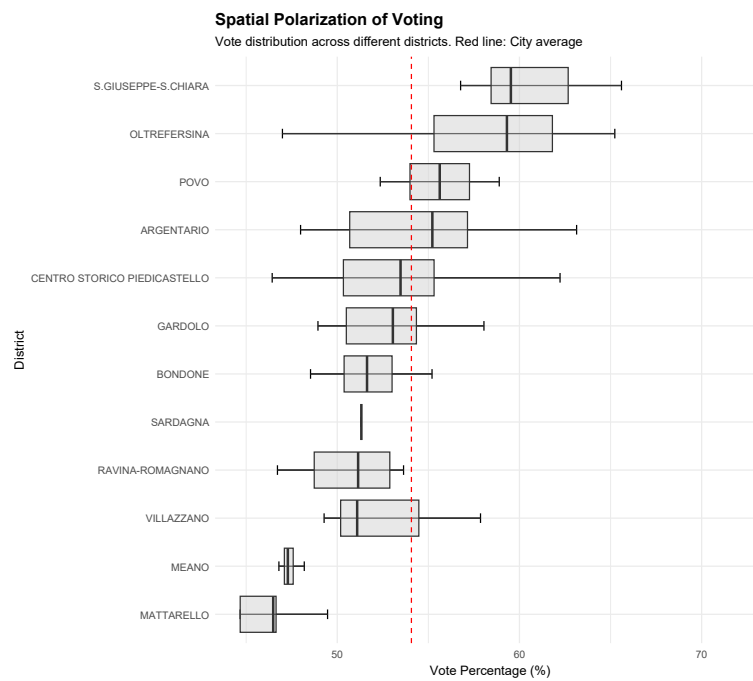


Figure 3: Spatial Polarization of Voting across Trento districts.

Table 1: Top 5 and Flop 5 Precincts by Vote Share.

Precinct ID	District	Vote Share (%)
<i>Top 5 Precincts</i>		
32	Oltrefersina	72.12
30	S. Giuseppe-S. Chiara	65.60
34	Oltrefersina	65.23
33	Oltrefersina	64.98
27	S. Giuseppe-S. Chiara	63.53
<i>Flop 5 Precincts</i>		
57	Mattarello	44.67
58	Mattarello	44.67
11	Centro Storico Piedicastello	46.43
55	Mattarello	46.48
56	Mattarello	46.64

- **Education:** A strong positive correlation (Pearson’s $r = 0.51$) is observed between the percentage of highly educated residents and the vote share, indicating that human capital is a primary driver of the candidate’s electoral base.
- **Urban Density:** Similarly, population density exhibits a strong positive trend ($r = 0.46$), reinforcing the hypothesis that the candidate performs better in highly urbanized environments.
- **Transit Accessibility:** The distance to the nearest public transport stop shows a moderate negative correlation ($r = -0.39$). Precincts located closer to transit networks tend to show higher support.
- **Immigration:** The presence of non-EU citizens shows a weaker, positive correlation ($r = 0.19$).

While these preliminary correlations are insightful, they do not account for the spatial dependence inherent in geographical data. A simple linear interpretation might be biased by the neighborhood effect. Therefore, the next section formally investigates the spatial autocorrelation of the electoral results.

4.2. Exploratory Spatial Data Analysis (ESDA)

To move beyond non-spatial descriptive statistics, we perform an Exploratory Spatial Data Analysis (ESDA) to formally assess the presence of spatial dependence in the electoral results.

Spatial Weights Matrix

The first step in the creation of a spatial model is to specify the spatial topology of the data generating process by defining a spatial weights matrix (W). Given the irregular shape and size of the electoral precincts in Trento, a contiguity-based approach could lead to isolated spatial units. Therefore, we adopted a k -nearest neighbors (KNN) criterion. We set $k = 4$ to ensure sufficient connectivity across the municipal territory while maintaining local relevance.

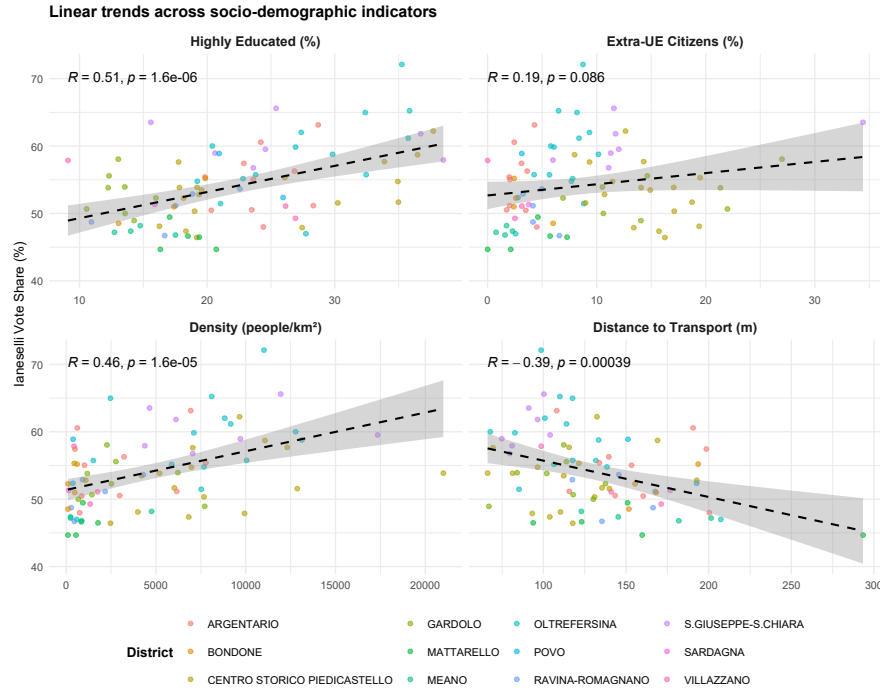


Figure 4: Bivariate scatterplots and linear trends for socio-demographic indicators.

The matrix was subsequently row-standardized to facilitate the interpretation of the spatial lag.

Global Spatial Autocorrelation

To test the hypothesis of spatial randomness, we computed the Global Moran's I statistic on the dependent variable (*Ianeselli's vote share*). The test yields a highly significant positive index of 0.601 ($p = 0.001$, based on 999 Monte Carlo permutations). This confirms a strong positive spatial autocorrelation: electoral support is not randomly distributed, but rather characterized by intense clustering, where precincts with high (or low) support for the incumbent mayor tend to be surrounded by precincts with similar voting behaviors.

The Moran scatterplot (Figure 5) provides a visual representation of this dependence, and shows that the vast majority of the observations fall into the High-High and Low-Low quadrants, confirming that spatial units are largely surrounded by neighbors with similar values. Furthermore, the plot highlights influential precincts (16, 18, 22, and 58).

Local Indicators of Spatial Association (LISA)

Since the global Moran's I statistic does not allow us to identify the specific locations of local spatial cluster, we rely on the Local Moran's I index to map the local patterns of spatial association.

Figure 6 compares the results using a standard significance threshold (raw $p < 0.05$) with those obtained by applying a strict Bonferroni correction for multiple testing. While the standard map (Figure 6a) reveals broader regional trends including a significant cluster in the northern areas, the Bonferroni correction (Figure 6b) acts as a severe spatial filter, isolating only the most statistically robust clusters.

This strict analysis identifies a clear geographic polarization: the *High-High* clusters (hotspots

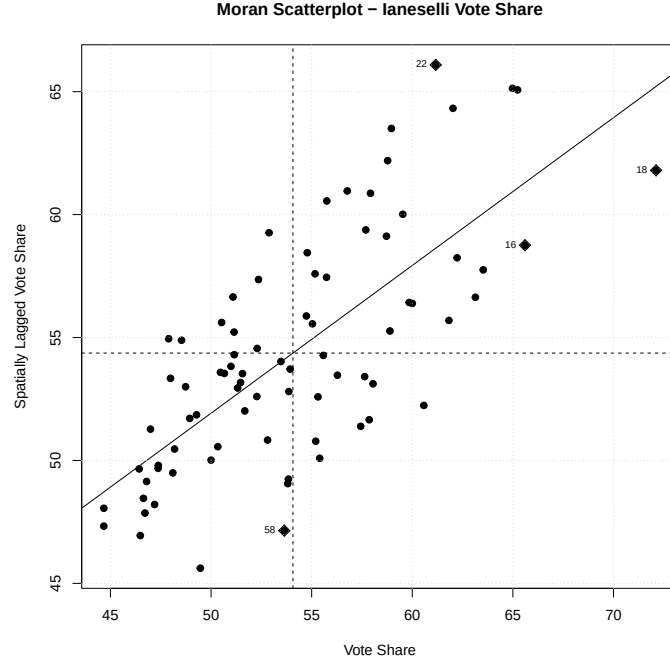


Figure 5: Moran Scatterplot for Ianeselli's Vote Share.

of electoral support) are solidly concentrated in the central and historical districts of the city. Conversely, the only *Low-Low* cluster (coldspot) is distinctly located in the southern periphery, specifically in the *Mattarello* district. This persistent geographic fracture demonstrates that local voting rates are heavily determined by the surrounding spatial context, validating the need for spatial regression modeling.

4.3. Non-Spatial Regression and Diagnostics

To establish a baseline for identifying the structural determinants of the progressive vote, we first estimated a standard Ordinary Least Squares (OLS) regression model. As reported in Table 3, the OLS results suggest that both the share of highly educated residents ($\beta = 0.335$, $p < 0.001$) and the proximity to public transit ($\beta = -0.035$, $p < 0.05$) are significant predictors of electoral support. Specifically, the transit coefficient indicates a substantial drop of 3.5 percentage points in the incumbent's vote share for every 100 meters of distance from the nearest stop.

However, traditional ecological regressions rely on the assumption of independent observations. Given the geographic nature of electoral data, this assumption is frequently violated. To test for omitted spatial dependence, we applied the Global Moran's I test to the studentized residuals of the OLS model. The test returned a highly significant positive index ($I = 0.343$, $p < 0.001$), confirming that the OLS estimates are biased and inconsistent due to unobserved spatial clustering.

To determine the precise nature of this spatial dependence, we performed the diagnostic Lagrange Multiplier (LM) tests (Table 2). Following the Elhorst (2010) model selection strategy, since both standard tests are highly significant, we must examine their robust counterparts.

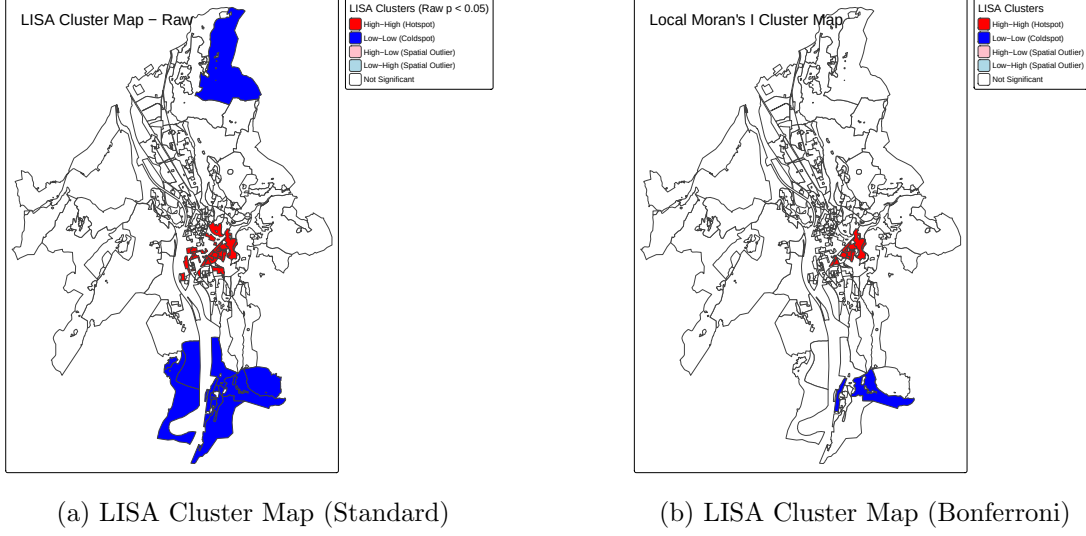


Figure 6: Comparison of Local Indicators of Spatial Association (LISA) for Ianeselli's vote share. The Bonferroni correction (right) filters out less robust associations, highlighting the most significant spatial clusters.

The robust LM-Lag test remains significant ($p = 0.010$), whereas the robust LM-Error fails to reject the null hypothesis ($p = 0.772$). This provides strong statistical evidence that the unobserved spatial clustering is of an autoregressive nature (spatial lag) rather than merely an error process.

Table 2: Lagrange Multiplier (LM) Diagnostics for Spatial Dependence

Test	Statistic	p-value
LM Error (Standard)	22.70	< 0.001
LM Lag (Standard)	29.12	< 0.001
Robust LM Error	0.08	0.772
Robust LM Lag	6.50	0.010

4.4. Spatial Model Selection and Estimation

Firstly, we estimated a Spatial Durbin Model (SDM), which accounts for spatial interactions in both the dependent variable and the independent covariates. To test whether this unrestricted model could be simplified, we performed Likelihood Ratio (LR) tests (ANOVA) comparing the SDM against the nested Spatial Autoregressive (SAR) and Spatial Error (SEM) models. The test comparing the SDM to the SAR model yielded a non-significant result ($LR = 6.20$, $p = 0.184$), leading us to accept the null hypothesis that the spatially lagged independent variables (WX) are not significantly different from zero. Consequently, the SAR model provides a more parsimonious and equally statistically robust fit for our data. The SAR model specification hypothesizes that the voting behavior in a specific precinct is directly influenced by the voting behavior of its neighboring precincts, capturing the substantive social contagion and neighborhood polarization effect.

Table 3: Regression Results: Spatial Models Comparison

	OLS	SAR	SEM	SDM
(Intercept)	50.096*** (3.180)	23.153*** (5.822)	50.841*** (2.911)	37.108*** (9.707)
highly_educated_pct	0.335*** (0.080)	0.165** (0.071)	0.193** (0.091)	0.051 (0.104)
extra_ue_pct	0.055 (0.091)	0.075 (0.074)	0.146* (0.082)	0.119 (0.081)
density	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
nearest_transport_distance_m	-0.035** (0.017)	-0.021 (0.014)	-0.020 (0.013)	-0.023* (0.014)
ρ (Spatial Lag)		0.535*** (0.105)		0.476*** (0.117)
λ (Spatial Error)			0.582*** (0.104)	
R^2	0.383	—	—	—
Num. obs.	80	80	80	80
Log Likelihood	-235.811	-221.241	-222.738	-218.138
AIC	476.257	456.482	459.476	458.275

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$

Note: Standard errors in parentheses.

4.5. Impact Analysis: Direct, Indirect, and Total Spillovers

In the SAR model, the spatial autoregressive coefficient ($\rho = 0.535$, $p < 0.001$) is highly significant and positive. This mathematically confirms the strong geographical polarization: over half of the electoral variance is associated with a social contagion multiplier effect.

Crucially, the inclusion of the spatial lag (ρ) structurally alters the model. Comparing the OLS and SAR columns in Table 3, we can observe that the public transit variable, which was significant in the non-spatial baseline, completely loses its statistical significance.

Moreover, in a spatial lag model, the raw β coefficients can no longer be interpreted as simple marginal effects. A change in a socio-demographic variable within a single precinct affects its own electoral outcome (Direct Impact), but through the spatial multiplier (ρ), it also influences neighboring precincts (Indirect Impact or Spillover), eventually feeding back to the origin. To correctly quantify and interpret these dynamics, we simulated the exact impacts using 1,000 iterations (Table 4).

The impact analysis reveals two critical findings:

1. **The Local Fallacy of Transit Distance:** The public transit variable, which appeared significant in the OLS regression, loses its statistical significance in the spatial framework (Total Impact $p = 0.151$). This highlights an important methodological pitfall: the distance from transit stops was merely capturing the underlying structural centrality of the precincts, acting as a spurious correlation. Once the model explicitly accounts for spatial clustering via the autoregressive lag, the localized infrastructure loses its direct

Table 4: Direct, Indirect (Spillover), and Total Impacts (SAR Model, $k = 4$)

Variable	Direct Impact	Indirect Impact	Total Impact	Total p -value
Highly Educated (%)	0.177**	0.178*	0.355**	0.019
Non-EU Residents (%)	0.080	0.080	0.160	0.341
Urban Density	0.000	0.000	0.000	0.728
Transit Distance (m)	-0.022	-0.022	-0.045	0.151

Note: Significance levels derived from simulated z -statistics ($R = 1000$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

explanatory power.

2. **The Educational Spillover:** The proportion of highly educated residents emerges as the only robust structural driver of the progressive vote. Notably, its effect is not strictly localized. An increase in the share of university graduates generates a significant Direct Impact ($p = 0.015$) on the local vote, but also produces a substantial Indirect Spillover effect ($p = 0.066$) on neighboring precincts. This suggests a cultural *halo effect*, where human capital fosters a progressive political climate that radiates across precinct boundaries.

Finally, to validate the structural integrity of the SAR model, we must verify that the inclusion of the spatial lag successfully absorbed the spatial dependence omitted by the baseline OLS. Figure 7 presents a visual comparison of the unstandardized residuals (prediction errors expressed in actual percentage points) for both models, plotted on an identical fixed scale.

The OLS error map clearly displays severe spatial clustering: the model systematically underestimates the vote share in the central districts (deep red clusters) and overestimates it in the periphery (blue clusters). Conversely, the SAR error map appears flattened, with residual errors distributed more randomly across the urban space and with lighter colors.

This visual intuition is mathematically confirmed by the Global Moran's I test performed on the SAR residuals. The test yields a virtually zero, non-significant result ($I = -0.039$, $p = 0.648$). We can conclude that the SAR framework successfully eliminated all unobserved spatial dependence, correcting the bias of the initial OLS approach and producing robust, unbiased estimates.

5. Conclusions

This study aimed to investigate the structural and spatial determinants of voting behavior during the May 2025 municipal elections in Trento, moving beyond traditional, spatially-blind ecological regressions. By integrating precinct-level electoral outcomes with ISTAT census indicators and open public transit data, we demonstrated that political consensus is deeply embedded in geographical and social contexts. The confirmation of the center-left incumbent mayor, Franco Ianeselli, secured at the first round with 54.65% of the votes amidst a historically low turnout (49.93%), provided a highly relevant case study to explore intra-urban polarization and localized discontent.

The empirical findings provide clear answers to our initial Research Questions (RQs). Addressing **RQ1** (Socio-demographic drivers), the analysis confirms that within the urban con-

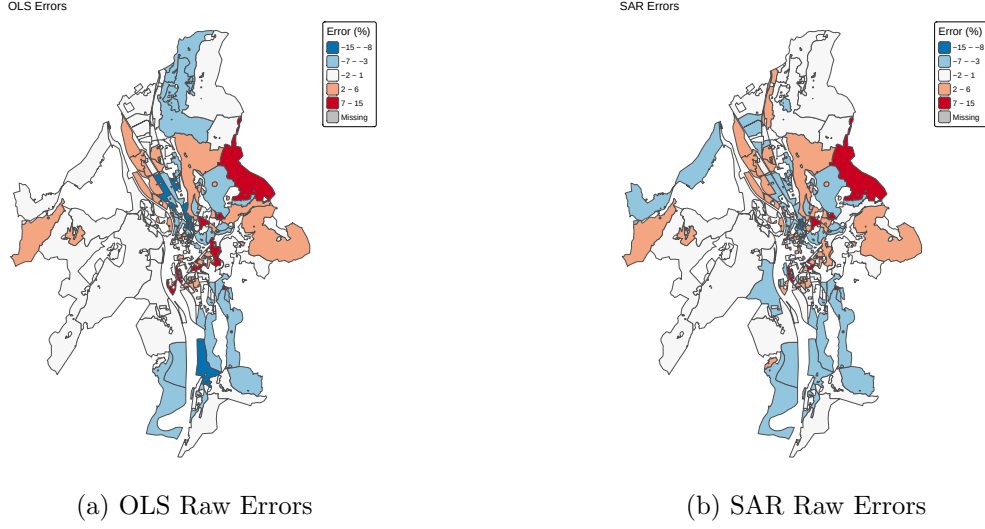


Figure 7: Comparison of Residual Errors (%). The models are plotted on a common scale. The SAR model (right) successfully breaks the spatial clusters of errors seen in the OLS (left).

text of Trento, human capital, measured by the share of highly educated residents, acts as the primary structural driver of the progressive vote.

Addressing **RQ2** (Spatial Contagion), the implementation of a Spatial Autoregressive (SAR) model revealed a highly significant positive spatial lag ($\rho = 0.535$, $p < 0.001$). This proves that electoral behavior across the city is not merely compositional (defined solely by who lives where), but strongly interdependent. Over half of the electoral variance is driven by spatial spillovers, indicating that human capital fosters a localized progressive climate that radiates across administrative boundaries via social interaction, acting as a cultural *halo effect*.

Crucially, answering **RQ3** (Methodological Bias), our framework exposed the severe limitations of standard linear models. In the baseline OLS regression, proximity to public transit appeared to be a highly significant predictor, suggesting a massive penalty of 3.5 percentage points for every 100 meters of distance from a stop. However, this effect completely vanished in the SAR specification. This drop confirms that the transit variable in the linear model was merely acting as a spurious correlation, capturing unobserved spatial centrality. Once the autoregressive lag explicitly accounted for spatial clustering, localized infrastructure lost all direct explanatory power, proving that non-spatial models introduce severe bias in political geography.

These methodological insights allow for a nuanced interpretation of localized urban grievances, specifically regarding the major construction sites for the *Bypass Ferroviario* (railway bypass). Descriptive and local spatial analyses (LISA) highlighted a sharp geographic fracture. In Trento Nord, while direct proximity to the excavation sites visibly penalized the incumbent, pushing bordering precincts (specifically sections 11, 12, and 14) below the absolute majority threshold, the broader electoral penalty was remarkably contained within the immediate perimeter of the works. Conversely, the southern suburb of Mattarello emerged as a distinct, statistically robust electoral coldspot, suffering a generalized and severe collapse in consensus across the entire district. While we must be cautious in defining higher education as an absolute cushion against dissent, the data suggests that the intersection of peripheral isolation and

intense construction disruption created an unmitigated NIMBY (Not In My Back Yard) effect in Mattarello. In the northern districts, the different socio-demographic fabric successfully buffered this dynamic, preventing a spatial spillover of discontent.

Finally, while the SAR framework successfully absorbed unobserved spatial dependence, as confirmed by the non-significant Moran's I on the residuals ($I = -0.039$, $p = 0.648$), two main methodological limitations must be acknowledged. First, due to the lack of granular, precinct-level data on abstentionism, it was impossible to perform spatial inference on voter turnout, leaving open the question of how low participation distorted the active electorate.

Second, our findings remain inherently subject to the Modifiable Areal Unit Problem (MAUP). This arises both from the arbitrary, static configuration of the municipal electoral precincts and from the spatial join methodology required to merge datasets: census blocks were assigned to electoral sections based on a majority rule (the mode of the residential street numbers falling within each boundary). Despite these constraints, this study reiterates the absolute necessity of integrating spatial econometrics into urban policy analysis to avoid spurious deductions and properly decode the geography of democracy.

6. Data and Code Availability

The complete analytical pipeline, from Python preprocessing to R geospatial analysis, is hosted at the following [GitHub Repository](#). An interactive version of the electoral maps, allowing for the dynamic exploration of voting patterns and socio-demographic indicators across Trento's precincts, is available as an [Interactive Web Map](#).

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A. Multicollinearity Diagnostics

To ensure the structural stability of our regression estimates, we verify the absence of severe multicollinearity among the independent covariates. Figure 8 displays the Pearson correlation matrix for all socio-demographic indicators and transit accessibility metrics. All pairwise correlations remain well below the conventional critical threshold of 0.80, confirming that the explanatory variables capture distinct dimensions of the urban fabric and do not inflate the standard errors of our spatial econometric models.

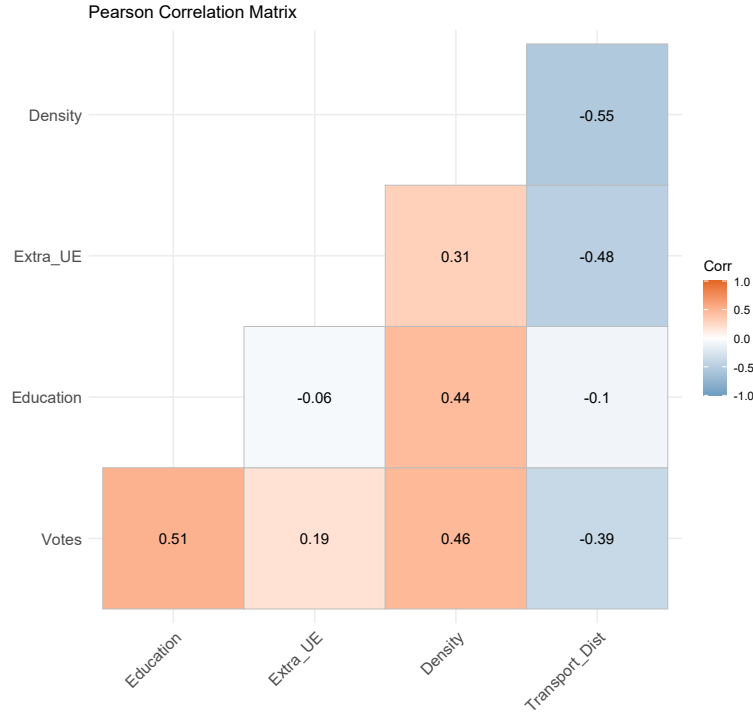


Figure 8: Pearson correlation matrix of the independent variables.

B. Robustness Checks: Sensitivity to the Spatial Weights Matrix

A fundamental methodological step in spatial econometrics is validating that the empirical findings are not driven by the arbitrary selection of the spatial weights matrix (W). In Section 3, we justified the use of a k -Nearest Neighbors topology with $k = 4$ to ensure complete connectivity across Trento's highly heterogeneous urban morphology. To formally test the sensitivity of our baseline Spatial Autoregressive (SAR) model, we re-estimated the baseline specification using denser neighborhood structures, specifically setting $k = 5$ and $k = 6$.

Table 5 presents the comparison of the estimated coefficients across the different spatial topologies. The point estimates and statistical significance of the parameters remain exceptionally stable. The spatial autoregressive coefficient (ρ) stays highly significant ($p < 0.001$) and hovers consistently between 0.53 and 0.57, confirming a robust spatial multiplier effect re-

ardless of the exact number of neighbors considered. Furthermore, the diagnostic Lagrange Multiplier test for residual autocorrelation remains non-significant across all specifications ($p > 0.10$), proving that the SAR framework successfully filters out spatial dependence in all configurations.

Table 5: Robustness Check: SAR Model Coefficients under alternative KNN topologies

Variable	SAR ($k = 4$, Baseline)	SAR ($k = 5$)	SAR ($k = 6$)
(Intercept)	23.153*** (5.822)	20.787*** (5.944)	21.159*** (6.252)
Highly Educated (%)	0.165* (0.071)	0.177* (0.071)	0.189** (0.073)
Non-EU Residents (%)	0.075 (0.074)	0.088 (0.074)	0.087 (0.075)
Urban Density	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
Transit Distance (m)	-0.021 (0.014)	-0.019 (0.013)	-0.021 (0.014)
ρ (Spatial Lag)	0.535*** (0.105)	0.568*** (0.108)	0.558*** (0.115)
AIC	456.48	455.37	457.36
Log Likelihood	-221.24	-220.69	-221.68
Num. Obs.	80	80	80
Residual Autocorr. LM p -val	0.212	0.179	0.218

Note: Asymptotic standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Crucially, because direct point estimates do not reflect true marginal responses in autoregressive models, we also computed the simulated Direct, Indirect (Spillover), and Total impacts for the alternative matrices (Table 6).

Table 6: Robustness Check: Comparison of Simulated Impacts across topologies

Variable	SAR ($k = 5$) Impacts			SAR ($k = 6$) Impacts		
	Direct	Indirect	Total	Direct	Indirect	Total
Highly Educated (%)	0.190***	0.220*	0.410**	0.200***	0.227*	0.428**
Non-EU Residents (%)	0.094	0.109	0.204	0.092	0.104	0.196
Urban Density	0.000	0.000	0.000	0.000	0.000	0.000
Transit Distance (m)	-0.020	-0.024	-0.044	-0.022	-0.025	-0.046

Note: Significance levels derived from simulated z -statistics ($R = 1000$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The impact sensitivity analysis fully validates our structural conclusions. The *Local Fallacy* regarding public transit accessibility holds true: once spatial spillovers are accounted for, the transit distance consistently shows no significant direct or total impact across all configurations. Conversely, the human capital variable (Highly Educated %) maintains highly significant direct effects alongside strong positive indirect spillovers. This confirms that the

identification of an educational *halo effect* driving progressive support is an intrinsic structural property of the electoral ecosystem, entirely independent of the spatial weights matrix design.